# KAPASA MAKASA UNIVERSITY

School of Applied Sciences and Open Learning | ICT Department

**Practical Test 2 — Advanced Integrated Assessment**

*Hello, and welcome to your Data Visualization practical. This paper is designed to genuinely stretch you — it asks you to make judgement calls the way a real analyst would, not just follow a checklist. Read each scenario carefully, work methodically, and don't panic if a step takes longer than expected: partial, well-reasoned progress is valued throughout the marking scheme. Take a breath, manage your time, and enjoy the challenge — we're rooting for you!*

**Course Name:** Data Visualization with Excel, Power BI and Tableau

**Duration:** 2 Hours

**Course Code:** DS 231

**Total Marks:** 100

## EXAMINATION INSTRUCTIONS

- **Format:** This is a strictly practical, hands-on examination spanning **three tools** — Microsoft Excel, Microsoft Power BI, and Tableau Desktop. You are required to demonstrate technical proficiency and independent analytical judgement across all three.
- **Resources:** You may only use the provided dataset(s) and the software installed on your workstation. No AI assistants, templates, or pre-built workbooks may be used.
- **Time Management:** You are not required to attempt the questions in order, but a suggested pacing of approximately 35–40 minutes per tool is recommended given the volume of work.
- **Submission:** At the end of the examination, you must submit **three files** to the following Google Drive link:

  *https://drive.google.com/drive/folders/2gT4mPz9RxKbAqLwN6vYsHc8jFdEo3ZtB*

  Ensure your files are named:
  - [YourStudentID]_Q1.xlsx
  - [YourStudentID]_Q2.pbix
  - [YourStudentID]_Q3.twbx
- **Integrity:** Independent work is required. Any detected file sharing, duplication, or identical calculated-field logic across submissions will result in an automatic zero for all parties involved.
- **Save Frequently:** It is your responsibility to save your work continuously in all three applications.

## Dataset Requirements

**Dataset Title:** Video Game Sales — Global Title-Level Sales by Platform, Genre & Publisher (1980–2020)

**Download Link:**

*https://www.kaggle.com/datasets/gregorut/videogamesales*

**Context:** This dataset tracks roughly 16,500 video game titles released worldwide since 1980, with regional and global sales figures in millions of units (NA_Sales, EU_Sales, JP_Sales, Other_Sales, Global_Sales), alongside Platform, Genre, Publisher, and release Year. It is a genuinely fun dataset to explore — Mario, Pokémon, GTA, and Call of Duty all make an appearance — but it is also a real, uncleaned export with plenty to catch out the unwary analyst:

- Missing **Year** values for a number of older or obscure titles, which cannot simply be dropped without quietly erasing part of gaming history from your trend analysis.
- Missing **Publisher** values for dozens of titles — treating these as "errors" to delete would understate the indie/small-publisher share of the market.
- Rounding drift between **Global_Sales** and the sum of **NA_Sales + EU_Sales + JP_Sales + Other_Sales** — the two don't always tie out exactly, and a trustworthy report needs to know by how much and where.
- A sprawling, inconsistent **Platform** field — *PS, PS2, PS3, PS4, PSP, PSV, XB, X360, XOne, Wii, WiiU, NES, SNES, N64, GC, GB, GBA, DS, 3DS* and more — that only becomes analytically useful once grouped into a console manufacturer/family.
- The same title appearing many times over — legitimately, as re-releases, remasters, or ports across platforms — which must be distinguished from genuine accidental duplicate rows.
- A long tail of extreme blockbuster outliers (think *Wii Sports* or *Grand Theft Auto V*) that are real, not data-entry errors, and need to be handled thoughtfully rather than simply deleted.

*Candidates are expected to make and justify their own reasonable data-cleaning decisions where the instructions do not spell out an exact rule — this is treated as part of the analytical skill being assessed, not an ambiguity to be resolved by guessing at a single "correct" answer.*

**Tips for Success**

- Skim all three questions first — some cleaning logic (missing values, platform/console standardisation, sales-consistency checks) is reused across Excel, Power BI, and Tableau, so decide your rules once and apply them consistently.
- Build your helper/flag columns before your visuals — a dashboard built on unreliable numbers will lose marks even if it looks polished.
- If you get stuck on one sub-task, move on and come back — every question is marked in discrete parts, so partial credit is always available.
- Name your worksheets, pages, and dashboards clearly — a marker who can navigate your file quickly is a marker who can find your best work.

## Question 1: Advanced Data Transformation & Interactive Dashboard  30 Marks

**Software:** Microsoft Excel (Power Query + PivotTables)

**Scenario:** You are the lead Data Analyst for a gaming analytics startup preparing a "40 Years of Gaming" report — part rigorous audit, part fun retrospective. Editorial has asked for a historically accurate, formula-driven reconstruction of the sales data — manual filtering or eyeballing rows to remove "bad" data is explicitly not acceptable, since the workbook must remain auditable and reproducible if the raw export is refreshed. They've also asked for a genuine data-science lens: which publishers actually move units, and which titles are quietly carrying the industry.

**Tasks:**

1. **Formula-Driven Cleaning Flags:** Import the dataset into Excel. Without deleting any raw rows, add calculated helper columns that use formulas (e.g. *ISBLANK, ROUND, ABS, SUMIF, TRIM, EXACT*) to flag: (a) rows with a missing **Year**, (b) rows with a missing **Publisher**, and (c) rows where **Global_Sales** does not reconcile with the sum of **NA_Sales + EU_Sales + JP_Sales + Other_Sales** beyond a small rounding tolerance (e.g. 0.01 million units).
2. **Calculated Columns & Outlier Detection:** Create a **Console Family** column that groups the raw **Platform** codes into manufacturer families (Nintendo, Sony, Microsoft, Sega, PC, Other) using nested *IF* or *IFS* logic. Then add an **Outlier Flag** column that uses a formula-based interquartile-range (IQR) rule — *QUARTILE.INC* and a 1.5×IQR fence — to flag blockbuster titles whose **Global_Sales** sit far outside the typical range, and briefly consider in your notes whether these should be excluded or celebrated.

3. **PivotTable-Driven Publisher Analysis & Pareto:** On a new worksheet, build a publisher-level table using PivotTables and formulas: *Recency* (years since each publisher's most recent release, relative to the dataset's latest year), *Frequency* (count of titles released), and *Total Sales* (sum of Global_Sales) — then use these to assign each publisher a **Publisher Tier** (High / Medium / Low) with a formula. Separately, build a **Pareto (80/20) analysis**: a PivotTable of Global_Sales by title with a running-total % column, revealing what share of titles generates 80% of all-time global sales.
4. **Executive Dashboard:** On a dedicated "Executive Dashboard" worksheet, using PivotCharts connected to your cleaned data, build: a line chart of Global_Sales by Year; a bar chart of the Top 10 Publishers by Total Sales; a chart showing the distribution of publishers across your Publisher Tiers; a slicer for Console Family and Genre connected across the charts; and conditional formatting that highlights any outlier or sales-mismatch rows still visible in your data view.

*Marks Allocation: Formula-Driven Cleaning Flags (7), Calculated Columns & Outlier Detection (7), Publisher & Pareto Pivot Analysis (9), Dashboard Design & Slicer Interactivity (7).*

**Stretch Challenge (optional, no extra marks — but excellent practice)**

Add a dynamic named range or Excel Table so your PivotTables auto-expand if the export is refreshed with newer titles, and see whether your Pareto analysis still roughly holds to an 80/20 split once outliers are excluded.

## Question 2: Enterprise BI Modelling with Time Intelligence 35 Marks

**Software:** Microsoft Power BI

**Scenario:** The Business Intelligence team wants a repeatable, governed model of the gaming sales data that survives a raw re-export, and that can explain not just total sales but how the industry is trending over time and which manufacturer ecosystems are winning. You are also given a small supplementary **Platform–Manufacturer reference table** (with a few of its own inconsistent platform codes) that must be merged in rather than typed in by hand.

**Tasks:**

5. **Power Query Transformations:** Import the dataset and load it into Power Query. Promote headers; correct the **Year** data type; add a conditional column **Sales Mismatch** flagging rows where Global_Sales doesn't reconcile with the sum of regional sales beyond a small tolerance; and instead of deleting rows with a blank **Publisher**, replace them with a flag value (e.g. "Unknown/Indie") so their sales are retained but reportable separately. Merge the Platform–Manufacturer reference table onto the main query and resolve any code mismatches so no platform is left unmatched (null Manufacturer).
6. **DAX Foundations:** Create explicit measures: **Total Global Sales**, **Total Titles** (distinct count of Name + Platform combinations), **Average Sales per Title**, regional mix measures **NA Share %** / **EU Share %** / **JP Share %** (each region's sales as a % of Global Sales), a **Hit Rate %** measure (titles with Global_Sales above 1 million units, as a % of all titles), and a time-intelligence measure **Sales YoY Growth %** using *SAMEPERIODLASTYEAR* or *DATEADD* on Year.
7. **Core Visualisations & Drillthrough:** On Page 1, build Card visuals for Total Global Sales, Hit Rate %, and Sales YoY Growth %; a line chart of Global Sales by Year with the prior-period trend overlaid for comparison; and a clustered bar chart of the Top 10 Publishers by Total Global Sales with a visual-level filter excluding one dominant manufacturer of your choice, to surface the rest of the competitive landscape. Add a drillthrough page that shows title-level detail when a publisher bar is right-clicked.

*Marks Allocation: Power Query Transformations & Reference Merge (13), DAX Measure & Time-Intelligence Accuracy (13), Visualisation, Filtering & Drillthrough (9).*

**Stretch Challenge (optional, no extra marks — but excellent practice)**

Add a measure that flags years where Hit Rate % improved versus the prior year, and format it conditionally on your line chart.

## Question 3: Integrated Publisher Analytics & Storytelling 35 Marks

**Software:** Tableau Desktop

**Scenario:** The CEO wants a publisher-centric story for the full history of the dataset: which publishers actually drive the industry, how concentrated that value is, and how the regional sales mix compares across markets. You must build this directly on top of the raw extract using the Level of Detail, data-reshaping, and parameter techniques from the lab programme, not on a pre-cleaned copy.

**Tasks:**

8. **Data Modelling & Reshaping:** Connect to the dataset. Use Tableau's **Pivot** (unpivot) feature on the data source page to reshape the wide **NA_Sales / EU_Sales / JP_Sales / Other_Sales** columns into two long-format fields, **Region** and **Regional Sales**, so that region becomes a proper dimension rather than four separate columns. Create a calculated field **Sales Consistency Check** comparing the summed Regional Sales against **Global_Sales** for each title.
9. **LOD Expressions, Sets & Parameters:** Build a **FIXED** LOD field **Publisher Lifetime Sales** (total Global_Sales per Publisher across the full dataset, independent of any filters in the view). Create an integer parameter **Top N Selector** and use it to drive a dynamic Top-N **Set** of publishers by Lifetime Sales. Using the FIXED LOD result, build a **Publisher Tier** field bucketing publishers into High / Medium / Low tiers.
10. **Executive Dashboard:** On a new dashboard, include: a treemap of Global_Sales by Genre and Console Family; a bar chart of the parameter-controlled Top-N publishers by Lifetime Sales; a stacked or side-by-side bar chart of the regional sales mix using your reshaped **Region** field; and a Year range filter covering the full history of the dataset. Add a dashboard filter action so that clicking a Genre in the treemap filters the publisher bar chart to that genre only.
11. **Data Storytelling & Polish:** Insert a text box directly onto the dashboard and write two distinct, actionable insights based entirely on the visuals you built (for example, about sales concentration among a handful of publishers, or a regional pattern in genre preference). Ensure a consistent colour theme, clear titles, and appropriate data labels throughout.

*Marks Allocation: Data Modelling & Reshaping (6), LOD/Parameter/Set Construction (12), Dashboard Layout & Interactivity (11), Analytical Insights & Polish (6).*

> **Stretch Challenge (optional, no extra marks — but excellent practice)**
>
> Add a trend line or forecast to your Global Sales by Year chart, and see if your two insights still hold once you account for it.

*That's the full paper. Take one more look at your file names and submission link before you finish up — well done for working through this, and good luck!*

**— End of Practical Test 2 —**